Fundamental Theorem of Calculus

Step-by-Step Worked Solutions

Problems 13–34

Section 1: Evaluating Integrals using FTC Part 1 (Problems 13–30)

13 $\displaystyle\int_{-2}^{1}(x^2-6x+12)\,dx$

Antiderivative:

$$\left[\frac{x^3}{3}-3x^2+12x\right]_{-2}^{1}$$

Evaluation:

$$x=1:\ \tfrac{1}{3}-3+12=\tfrac{28}{3}$$
$$x=-2:\ -\tfrac{8}{3}-12-24=-\tfrac{116}{3}$$
$$\text{Total: } \tfrac{28}{3}+\tfrac{116}{3}=\tfrac{144}{3}=\boxed{48}$$

14 $\displaystyle\int_{-1}^{2}4x(1-x^2)\,dx=\int_{-1}^{2}(4x-4x^3)\,dx$

$$\left[2x^2-x^4\right]_{-1}^{2}=(8-16)-(2-1)=-8-1=\boxed{-9}$$

15 $\displaystyle\int_{1}^{4}\frac{4}{x^2}\,dx=\int_{1}^{4}4x^{-2}\,dx$

$$\left[-\frac{4}{x}\right]_{1}^{4}=-1-(-4)=\boxed{3}$$

16 $\displaystyle\int_{1}^{2}\frac{1}{x^6}\,dx=\int_{1}^{2}x^{-6}\,dx$

$$\left[-\frac{1}{5x^5}\right]_{1}^{2}=-\frac{1}{160}+\frac{1}{5}=\boxed{\frac{31}{160}}$$

17 $\displaystyle\int_{4}^{9}2x\sqrt{x}\,dx=\int_{4}^{9}2x^{3/2}\,dx$

$$\left[2\cdot\tfrac{2}{5}x^{5/2}\right]_{4}^{9}=\left[\tfrac{4}{5}x^{5/2}\right]_{4}^{9}=\tfrac{4}{5}(243-32)=\tfrac{4}{5}(211)=\boxed{\frac{844}{5}}$$

18 $\displaystyle\int_{1}^{4}\frac{1}{x\sqrt{x}}\,dx=\int_{1}^{4}x^{-3/2}\,dx$

$$\left[-\frac{2}{\sqrt{x}}\right]_{1}^{4}=-\frac{2}{2}-\left(-\frac{2}{1}\right)=-1+2=\boxed{1}$$

19 $\displaystyle\int_{-\pi/2}^{\pi/2} \sin\theta \, d\theta$

$$[-\cos\theta]_{-\pi/2}^{\pi/2} = -\cos\tfrac{\pi}{2} + \cos\left(-\tfrac{\pi}{2}\right) = 0 - 0 = \boxed{0}$$

20 $\displaystyle\int_{0}^{\pi/4} \sec^2\theta \, d\theta$

$$[\tan\theta]_{0}^{\pi/4} = 1 - 0 = \boxed{1}$$

21 $\displaystyle\int_{-\pi/4}^{\pi/4} \cos x \, dx$

$$[\sin x]_{-\pi/4}^{\pi/4} = \tfrac{\sqrt{2}}{2} - \left(-\tfrac{\sqrt{2}}{2}\right) = \boxed{\sqrt{2}}$$

22 $\displaystyle\int_{0}^{\pi/3} (2x - \sec x \tan x) \, dx$

$$\left[x^2 - \sec x\right]_{0}^{\pi/3} = \left(\tfrac{\pi^2}{9} - 2\right) - (0 - 1) = \boxed{\frac{\pi^2}{9} - 1}$$

23 $\displaystyle\int_{\ln 2}^{3} 5e^x \, dx$

$$[5e^x]_{\ln 2}^{3} = 5e^3 - 5e^{\ln 2} = 5e^3 - 10 \quad \Rightarrow \quad \boxed{5e^3 - 10}$$

24 $\displaystyle\int_{1/2}^{1} \frac{1}{2x} \, dx = \frac{1}{2}\int_{1/2}^{1} \frac{1}{x} \, dx$

$$\left[\tfrac{1}{2}\ln|x|\right]_{1/2}^{1} = \tfrac{1}{2}\ln 1 - \tfrac{1}{2}\ln\tfrac{1}{2} = 0 - \tfrac{1}{2}(-\ln 2) = \boxed{\frac{\ln 2}{2}}$$

25 $\displaystyle\int_{0}^{1/\sqrt{2}} \frac{dx}{\sqrt{1 - x^2}}$

$$[\arcsin x]_{0}^{1/\sqrt{2}} = \tfrac{\pi}{4} - 0 = \boxed{\frac{\pi}{4}}$$

26 $\displaystyle\int_{-1}^{1} \frac{dx}{1 + x^2}$

$$[\arctan x]_{-1}^{1} = \tfrac{\pi}{4} - \left(-\tfrac{\pi}{4}\right) = \boxed{\frac{\pi}{2}}$$

27 $\displaystyle\int_{\sqrt{2}}^{2} \frac{dx}{x\sqrt{x^2 - 1}}$

$$[\operatorname{arcsec} x]_{\sqrt{2}}^{2} = \operatorname{arcsec} 2 - \operatorname{arcsec}\sqrt{2} = \tfrac{\pi}{3} - \tfrac{\pi}{4} = \boxed{\frac{\pi}{12}}$$

28 $\displaystyle\int_{-\sqrt{2}}^{-2/\sqrt{3}} \frac{dx}{x\sqrt{x^2-1}}$

Antiderivative (valid for $x<0$ as well):

$$\left[\operatorname{arcsec}|x|\right]_{-\sqrt{2}}^{-2/\sqrt{3}}$$

Evaluation:

$$\operatorname{arcsec}\left(\tfrac{2}{\sqrt{3}}\right)-\operatorname{arcsec}\sqrt{2}=\tfrac{\pi}{6}-\tfrac{\pi}{4}=\boxed{-\frac{\pi}{12}}$$

29 $\displaystyle\int_1^4\left(\frac{1}{\sqrt{t}}-3\sqrt{t}\right)dt=\int_1^4\left(t^{-1/2}-3t^{1/2}\right)dt$

$$\left[2\sqrt{t}-2t^{3/2}\right]_1^4=(4-16)-(2-2)=\boxed{-12}$$

30 $\displaystyle\int_{\pi/6}^{\pi/2}\left(x+\frac{2}{\sin^2 x}\right)dx=\int_{\pi/6}^{\pi/2}(x+2\csc^2 x)\,dx$

Antiderivative:

$$\left[\frac{x^2}{2}-2\cot x\right]_{\pi/6}^{\pi/2}$$

Evaluation:

$$x=\tfrac{\pi}{2}:\ \tfrac{\pi^2}{8}-2\cot\tfrac{\pi}{2}=\tfrac{\pi^2}{8}$$
$$x=\tfrac{\pi}{6}:\ \tfrac{\pi^2}{72}-2\cot\tfrac{\pi}{6}=\tfrac{\pi^2}{72}-2\sqrt{3}$$
$$\text{Difference: }\left(\tfrac{\pi^2}{8}-\tfrac{\pi^2}{72}\right)+2\sqrt{3}=\boxed{\frac{\pi^2}{9}+2\sqrt{3}}$$

Section 2: Integrals Involving Absolute Values (Problems 31–32)

31(a) $\displaystyle\int_{-1}^{1}|2x-1|\,dx$

Root: $2x-1=0\implies x=\tfrac{1}{2}$. *Split:*

$$\int_{-1}^{1/2}(1-2x)\,dx+\int_{1/2}^{1}(2x-1)\,dx=\left[x-x^2\right]_{-1}^{1/2}+\left[x^2-x\right]_{1/2}^{1}$$

$$=\left(\tfrac{1}{4}-(-2)\right)+\left(0-\left(-\tfrac{1}{4}\right)\right)=\tfrac{9}{4}+\tfrac{1}{4}=\boxed{\frac{5}{2}}$$

31(b) $\displaystyle\int_0^{3\pi/4}|\cos x|\,dx$

Root: $\cos x=0\implies x=\tfrac{\pi}{2}$. *Split:*

$$\int_0^{\pi/2} \cos x\,dx + \int_{\pi/2}^{3\pi/4} (-\cos x)\,dx = [\sin x]_0^{\pi/2} + [-\sin x]_{\pi/2}^{3\pi/4}$$

$$= (1-0) + \left(-\tfrac{\sqrt{2}}{2} + 1\right) = \boxed{2 - \frac{\sqrt{2}}{2}}$$

32(a) $\displaystyle\int_{-1}^{2} \sqrt{2+|x|}\,dx$

Split at $x = 0$*:*

$$\text{On } [-1,0]: \int_{-1}^{0} \sqrt{2-x}\,dx = \left[-\tfrac{2}{3}(2-x)^{3/2}\right]_{-1}^{0} = 2\sqrt{3} - \tfrac{4\sqrt{2}}{3}$$

$$\text{On } [0,2]: \int_{0}^{2} \sqrt{2+x}\,dx = \left[\tfrac{2}{3}(2+x)^{3/2}\right]_{0}^{2} = \tfrac{16}{3} - \tfrac{4\sqrt{2}}{3}$$

$$\text{Total: } \boxed{\frac{16}{3} + 2\sqrt{3} - \frac{8\sqrt{2}}{3}}$$

32(b) $\displaystyle\int_0^{\pi/2} \left|\tfrac{1}{2} - \cos x\right| dx$

Root: $\cos x = \frac{1}{2} \implies x = \frac{\pi}{3}$*. Split:*

$$\text{On } [0, \tfrac{\pi}{3}]: \int_0^{\pi/3} \left(\cos x - \tfrac{1}{2}\right) dx = \left[\sin x - \tfrac{x}{2}\right]_0^{\pi/3} = \tfrac{\sqrt{3}}{2} - \tfrac{\pi}{6}$$

$$\text{On } [\tfrac{\pi}{3}, \tfrac{\pi}{2}]: \int_{\pi/3}^{\pi/2} \left(\tfrac{1}{2} - \cos x\right) dx = \left[\tfrac{x}{2} - \sin x\right]_{\pi/3}^{\pi/2} = \tfrac{\pi}{12} - 1 + \tfrac{\sqrt{3}}{2}$$

$$\text{Total: } \boxed{\sqrt{3} - \frac{\pi}{12} - 1}$$

Section 3: Chapter 5 Absolute Value Integrals (Problems 33–34)

33(a) $\displaystyle\int_{-1}^{1} |e^x - 1|\,dx$

Root: $e^x - 1 = 0 \implies x = 0$*. Split:*

$$\int_{-1}^{0} (1 - e^x)\,dx + \int_0^1 (e^x - 1)\,dx = [x - e^x]_{-1}^{0} + [e^x - x]_0^1 = e^{-1} + (e-2) = \boxed{e + e^{-1} - 2}$$

33(b) $\displaystyle\int_1^4 \frac{|2-x|}{x}\,dx$

Root: $2 - x = 0 \implies x = 2$*. Split:*

$$\text{On } [1,2]: \int_1^2 \left(\tfrac{2}{x} - 1\right) dx = [2\ln x - x]_1^2 = 2\ln 2 - 1$$
$$\text{On } [2,4]: \int_2^4 \left(1 - \tfrac{2}{x}\right) dx = [x - 2\ln x]_2^4 = 2 - 2\ln 2$$
$$\text{Total: } (2\ln 2 - 1) + (2 - 2\ln 2) = \boxed{1}$$

34(a) $\displaystyle\int_{-3}^{3} \left| x^2 - 1 - \frac{15}{x^2+1} \right| dx$

Integrand: $\dfrac{x^4-16}{x^2+1}$*, with roots at* $x = \pm 2$*. It is an even function, so evaluate* $2\int_0^3$*:*

$$\text{On } [0,2]: \int_0^2 \frac{16 - x^4}{x^2+1}\, dx = \left[-\tfrac{x^3}{3} + x + 15\arctan x\right]_0^2 = 15\arctan 2 - \tfrac{2}{3}$$
$$\text{On } [2,3]: \int_2^3 \frac{x^4 - 16}{x^2+1}\, dx = \left[\tfrac{x^3}{3} - x - 15\arctan x\right]_2^3 = \tfrac{16}{3} - 15\arctan 3 + 15\arctan 2$$
$$\text{Total: } 2\left(30\arctan 2 - 15\arctan 3 + \tfrac{14}{3}\right) = \boxed{\frac{28}{3} + 60\arctan 2 - 30\arctan 3}$$

34(b) $\displaystyle\int_0^{\sqrt{3}/2} \left| \frac{1}{\sqrt{1-x^2}} - \sqrt{2} \right| dx$

Root: $\dfrac{1}{\sqrt{1-x^2}} = \sqrt{2} \implies x = \dfrac{1}{\sqrt{2}}$*. Split:*

$$\text{On } [0, \tfrac{1}{\sqrt{2}}]: \int_0^{1/\sqrt{2}} \left(\sqrt{2} - \tfrac{1}{\sqrt{1-x^2}}\right) dx = \left[\sqrt{2}x - \arcsin x\right]_0^{1/\sqrt{2}} = 1 - \tfrac{\pi}{4}$$
$$\text{On } [\tfrac{1}{\sqrt{2}}, \tfrac{\sqrt{3}}{2}]: \int_{1/\sqrt{2}}^{\sqrt{3}/2} \left(\tfrac{1}{\sqrt{1-x^2}} - \sqrt{2}\right) dx = \left[\arcsin x - \sqrt{2}x\right]_{1/\sqrt{2}}^{\sqrt{3}/2} = \tfrac{\pi}{12} + 1 - \tfrac{\sqrt{6}}{2}$$
$$\text{Total: } \boxed{2 - \frac{\pi}{6} - \frac{\sqrt{6}}{2}}$$